

Windows Security Monitoring and System Analysis

Practical Project Report

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Objective

To perform basic monitoring of a Windows system by reviewing resource usage, active processes, services, event logs, security status, and network configuration.

Tools / Environment

- Windows 10/11
- Task Manager
- Resource Monitor
- Event Viewer
- Windows Security / Microsoft Defender
- Command Prompt and PowerShell

Lab Setup

I used my own Windows computer. Before starting, I closed unrelated applications where possible and then checked normal system activity using built-in Windows tools.

Practical Steps Performed

- Opened Task Manager and reviewed running processes, CPU utilization, memory usage, disk activity, and startup applications.
- Opened Resource Monitor to inspect CPU, memory, disk, and network activity in more detail.
- Checked Windows Security to review Microsoft Defender protection status.
- Opened Event Viewer and reviewed recent entries under System, Application, and Security logs.
- Ran system information and network commands in Command Prompt.
- Used PowerShell to view a small list of services.

Commands / Filters Used

```
whoami
systeminfo
ipconfig /all
tasklist
Get-Service | Select-Object -First 20
```

Observations

- Task Manager displayed normal background processes along with CPU and memory consumption.
- Resource Monitor provided process-wise details for disk and network use.
- Microsoft Defender status was reviewed as part of the basic security check.
- Event Viewer showed informational, warning, and/or error entries that can be used during troubleshooting.
- The ipconfig output showed adapter configuration, IPv4 address, default gateway, and DNS information.

Result

I completed a basic Windows health and security monitoring review using built-in tools. The exercise showed how system resources, logs, services, and security controls can be checked during troubleshooting.

What I Learned

- How to use Task Manager and Resource Monitor for performance troubleshooting.
- How Event Viewer supports basic investigation of system and security events.
- How to collect basic system and network information using Windows commands.

Screenshots to Add

- Task Manager Processes or Performance tab.
- Resource Monitor overview.
- Event Viewer log entry with sensitive data hidden.
- Windows Security / Defender status.
- Command Prompt output for ipconfig /all.

Ethical Note: This work was performed only in a personal/authorized lab environment. Sensitive information such as passwords, private IP details, MAC addresses, and personal data should be hidden before sharing screenshots or uploading material to GitHub.